

Pattabiraman Venkatesan

Lead Software Engineer

CONTACT

+91 9952682214
pattabi.rmn@gmail.com
linkedin.com/in/pattabiraman-venkatesan
Chennai, Tamil Nadu

EXPERIENCE

13+ Years

Enterprise Software Engineering

LANGUAGES

- Java
- Groovy
- JavaScript
- PowerShell

FRAMEWORKS

- Spring Boot
- Spring MVC
- Microservices
- REST APIs
- React / jQuery

AI & AUTOMATION

- GitHub Copilot
- GHCP SpecKit
- Spec-Driven Dev
- Agentic Testing
- Auto-Test Agent

DEVOPS & CLOUD

- Docker
- Kubernetes
- Azure CLI
- Azure Pipelines
- CI/CD Automation

SECURITY

- Veracode
- BlackDuck
- DevSecOps
- OSS Remediation

PROFESSIONAL SUMMARY

Lead Software Engineer with 13+ years specializing in Java/Spring Boot, cloud-native architecture, and DevSecOps — with a growing focus on AI-augmented engineering and agentic workflow design. Proven track record: reduced deployment effort by 60%, slashed client upgrade time by 75%, and improved security posture by 50%. Designed and built an agentic Auto-Test Agent with a self-improving feedback loop, now adopted organization-wide. Experienced in containerization, Azure migrations, CI/CD design, spec-driven development, and leading cross-functional teams.

PROFESSIONAL EXPERIENCE

Lead Product Software Engineer

Apr 2019 – Present

Wolters Kluwer ELM Solutions

AI-Augmented Engineering & Agentic Automation

- Pioneered Spec-Driven Development using GitHub Copilot SpecKit — authoring specs that delivered 20+ bug fixes across 4 production specs at **4x the speed** of conventional workflows.
- Architected a Groovy Unit Test framework for legacy codebases: automated extraction of Groovy code from XML archives, resolved runtime dependencies, executed tests, and reported code coverage — fully operational locally and in CI via **Copilot-powered PowerShell**.
- Created a reusable UT Skill (AI pattern library) enabling consistent, context-aware unit test generation — adopted across teams as part of the organization's Clean-as-You-Code initiative.
- Designed an Auto-Test Agent implementing a **generate-learn-fix feedback loop**: identifies least-covered classes, generates tests to a configurable 80% threshold, auto-repairs failures, and feeds learnings back into the skill — making the agent progressively more accurate.

Platform Engineering & Cloud

- Containerized the core product and led Azure cloud migration, **reducing deployment effort by 60%** across dev and production environments.
- Built an automated client upgrade pipeline, eliminating manual steps and **reducing upgrade time per environment by 75%** — now the standard delivery mechanism for all client onboarding.
- Led OSS security remediation via Veracode and BlackDuck — resolving critical CVEs and **improving overall security posture by 50%** across the product suite.
- Engineered and automated CI/CD pipelines, accelerating release cycles and simplifying developer onboarding into containerized environments.

Architecture & Delivery

- Architected Timekeeper Benchmark Rates microservice POC — defined RESTful API contracts and evaluated scalability under realistic load profiles.
- Led high-complexity framework upgrades (Drools, jQuery) within committed timelines, ensuring clients receive up-to-date, secure product versions.
- Mentored junior and mid-level engineers; received multiple Quarterly Star Awards for coaching and skill uplift across the team.

Senior Software Engineer

May 2012 – Mar 2019

Cognizant Technology Solutions

- Reduced LexisAdvance bug backlog from **500+ to under 50 in 9 months** via redesigned triage and systematic root-cause analysis.
- Co-built a scalable MarkLogic Admin Tool used across dev and production environments, refactoring core logic for significant performance and reliability gains.
- Designed and implemented a relevance evaluation framework from scratch (DCG, Jaccard Index, RBO, SERP Analytics) — adopted as a reusable internal quality benchmarking standard.
- Piloted a document rating tool that improved milestone delivery speed by 30%.

TOOLS

- MS SQL Server
- Git / Jira
- Confluence
- JUnit / Playwright

EDUCATION

B.Tech — Info Tech

VIT University

Vellore, TN | 2012

GPA: 7.85 / 10

AWARDS

- Best New Entrant — WK ELM
- Best New Entrant — CTS
- Quarterly Star (Mentoring)
- Best Bug Buster — CTS
- Gamified Learning Impact

KEY ACHIEVEMENTS

- 🏆 Best New Entrant, Engineering — WK ELM Solutions (2018–19)
- 🏆 Best New Entrant — Cognizant Technology Solutions (2012–13)
- 🏆 Quarterly Star Awards for Mentoring — WK ELM Solutions (multiple years)
- 🏆 Best Bug Buster — Cognizant Technology Solutions (multiple, 2014–15)
- 🏆 Designed gamified learning program → 50% increase in skill adoption & engagement
- 🏆 Auto-Test Agent adopted org-wide as part of Clean-as-You-Code engineering standard